Total Points: [10] [U1HW5 - W7 - Identifying Variables]

1. If a car drives faster, then it will use more gasoline:

- dependent; gasoline consumption (miles per gallon), independent; speed of the car (miles per hour)
- dependent; type of car, independent; road conditions
- dependent; speed of the car (miles per hour), independent; gasoline consumption (miles per gallon)
- dependent; the car, independent; the gasoline
- dependent; the gasoline, independent; the car
- dependent; road conditions, independent; type of car

Answer:

2. If a student studies for a test for a longer period of time, then they will get a better grade on the test:

- dependent; the student, independent; the test
- dependent; the test, independent; the student
- dependent; test grade (percentage), independent; amount of time studying (hours)
- dependent; amount of time studying (hours), independent; test grade (percentage)
- dependent; difficulty of the test, independent; student's intelligence
- dependent; student's intelligence, independent; difficulty of the test

Answer:

3. If you drink more water, then you will feel more hydrated:

- dependent; time of day, independent; type of water
- dependent; type of water, independent; time of day
- dependent; the water, independent; the person
- dependent; feeling of hydration (scale 1-10), independent; amount of water consumed (ounces)
- dependent; amount of water consumed (ounces), independent; feeling of hydration (scale 1-10)
- dependent; the person, independent; the water

Answer:

4. If a person exercises more frequently, then they will have a lower resting heart rate:

- dependent; the exercise, independent; the person
- dependent; the person, independent; the exercise
- dependent; frequency of exercise (sessions per week), independent; resting heart rate (beats per minute)

• dependent; type of exercise, independent; duration of exercise

• dependent; resting heart rate (beats per minute), independent; frequency of exercise (sessions per week)

• dependent; duration of exercise, independent; type of exercise

Answer:

5. If you listen to loud music, then your hearing can be damaged:

• dependent; the music, independent; the person

• dependent; level of hearing damage (audiologist's report), independent; volume of music (decibels)

• dependent; type of music, independent; duration of listening

• dependent; volume of music (decibels), independent; level of hearing damage (audiologist's report)

• dependent; duration of listening, independent; type of music

• dependent; the person, independent; the music

Answer:

6. If you get more sleep, then you will feel more alert during the day:

• dependent; level of alertness (self-reported scale), independent; amount of sleep (hours)

• dependent; time of waking up, independent; quality of sleep

• dependent; amount of sleep (hours), independent; level of alertness (self-reported scale)

• dependent; the person, independent; the sleep

• dependent; quality of sleep, independent; time of waking up

• dependent; the sleep, independent; the person

Answer:

7. If the temperature of the water is higher, then the sugar will dissolve faster:

• dependent; amount of sugar, independent; amount of water

• dependent; temperature of the water (°C), independent; time it takes for sugar to dissolve (seconds)

• dependent; the water, independent; the sugar

• dependent; amount of water, independent; amount of sugar

• dependent; time it takes for sugar to dissolve (seconds), independent; temperature of the water (°C)

• dependent; the sugar, independent; the water

Answer:

8. If a plant is exposed to more sunlight, then it will grow taller:

- dependent; the plant, independent; the sun
- dependent; plant height (cm), independent; amount of sunlight exposure (hours/day)
- dependent; soil type, independent; type of plant
- dependent; the sun, independent; the plant
- dependent; amount of sunlight exposure (hours/day), independent; plant height (cm)
- dependent; type of plant, independent; soil type

Answer:

9. If you use a stronger fertilizer, then your plants will produce more fruit:

- dependent; the fertilizer, independent; the plant
- dependent; the plant, independent; the fertilizer
- dependent; strength of fertilizer, independent; amount of fruit produced (number or weight)
- dependent; amount of fruit produced (number or weight), independent; strength of fertilizer
- dependent; type of plant, independent; amount of water
- dependent; amount of water, independent; type of plant

Answer:

10. If you eat more vegetables, then you will have more energy:

- dependent; cooking method, independent; type of vegetable
- dependent; the vegetables, independent; the person
- dependent; amount of vegetables eaten (grams), independent; energy level (self-reported scale)
- dependent; energy level (self-reported scale), independent; amount of vegetables eaten (grams)
- dependent; type of vegetable, independent; cooking method
- dependent; the person, independent; the vegetables

Answer: